

CSN205 Static Networks Assignment 1



Seneca college wants to set a new building for a new program, Artificial Intelligence. The building will have a Program coordinator, Faculties, Registrar, Secretaries, and Students.

You are an IT Technician, have been assigned to set the switches for this building. There are three switches as access switches to distribute the internet for all staff and students. There are two core switches for redundancy. The rooms have VoIP phones.

Please create the network Topology (next page) in a Packet Tracer file following the next steps.

Step 1 (20 Points):

- Set local username and password (10 Points)
 - o Username: [put your Seneca username] & Password: CSN205 (5 Points)
 - o Make sure the password associated with the username is encrypted (5 Points)
- Set authentication access to all switches as console and VTY access (10 Points)
- Set hostname for each Switch (SW1 to SW5)

Step 2 (30 Points): [Hints: chapter 8, example 8-8](#)

The access and core switches are supposed to be set for VLANs as in the following table:

- Set all VLANs with respected names in all 5 Switches (10 Points)
- Assign the interfaces to the respected VLANs only in all 3 Access Switches (20 Points)

VLAN	VLAN Name	Interfaces assigned to VLAN (only 3 Access Switches)
10	[put your first name]	Fa0/1-fa0/2
15	Faculty	Fa0/3-fa0/7
20	Registrar	Fa0/8-fa0/9
25	Secretary	Fa0/10-fa0/13
30	Voice	Fa0/1-fa0/13
35	Students	Fa0/14-fa0/24(both 3 Access & 2 Core Switches)

Step 3 (10 Points): [Hint: Chapter 8: VLAN Trunking Configuration](#)

The access switches connect to each core switch. There is a connection between the two core switches. All these connections between all the switches must be set as **Trunk** and pass all VLANs between them.

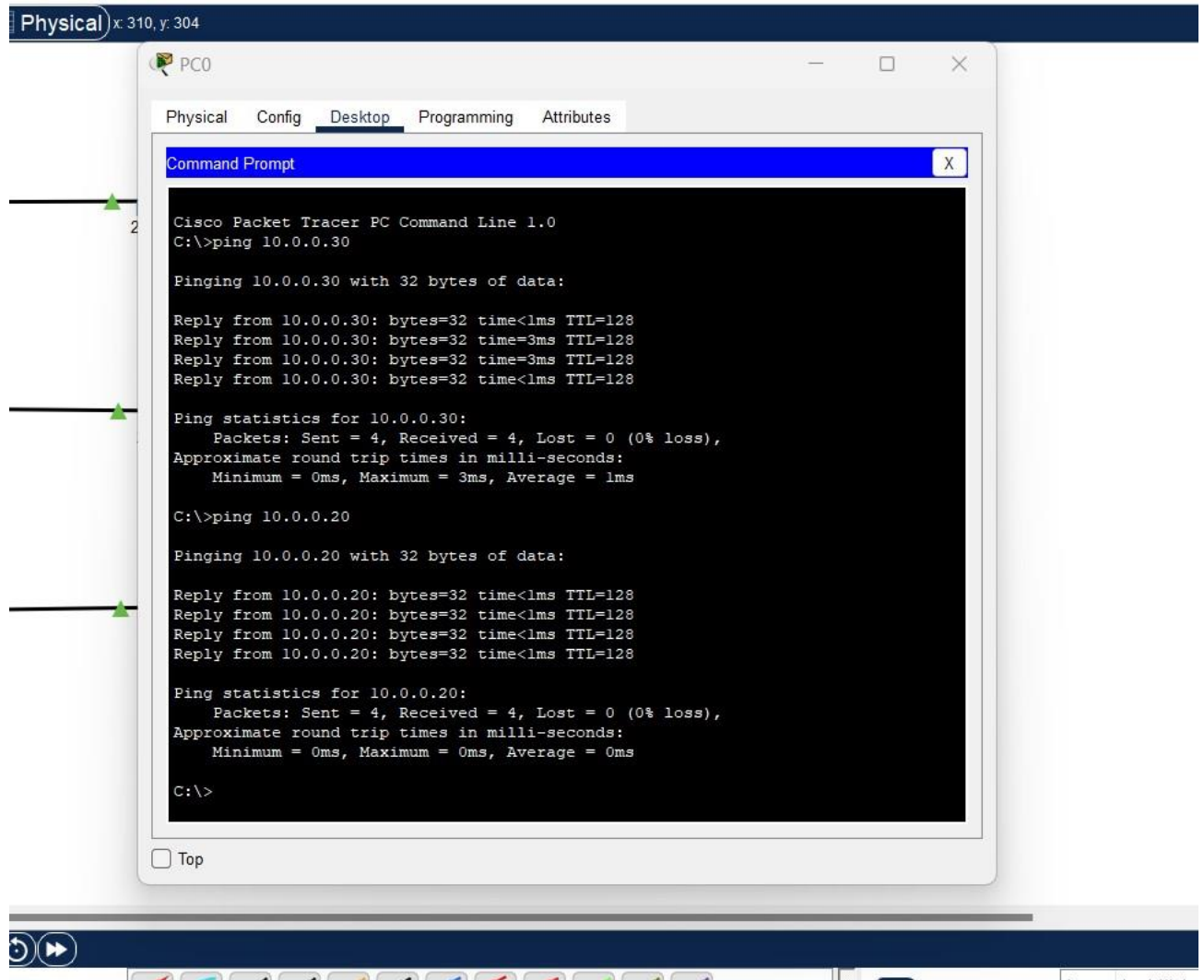
- According to the diagram (last page) use the Gi 0/1 & Gi 0/2 in each Access Switches (5 Points)

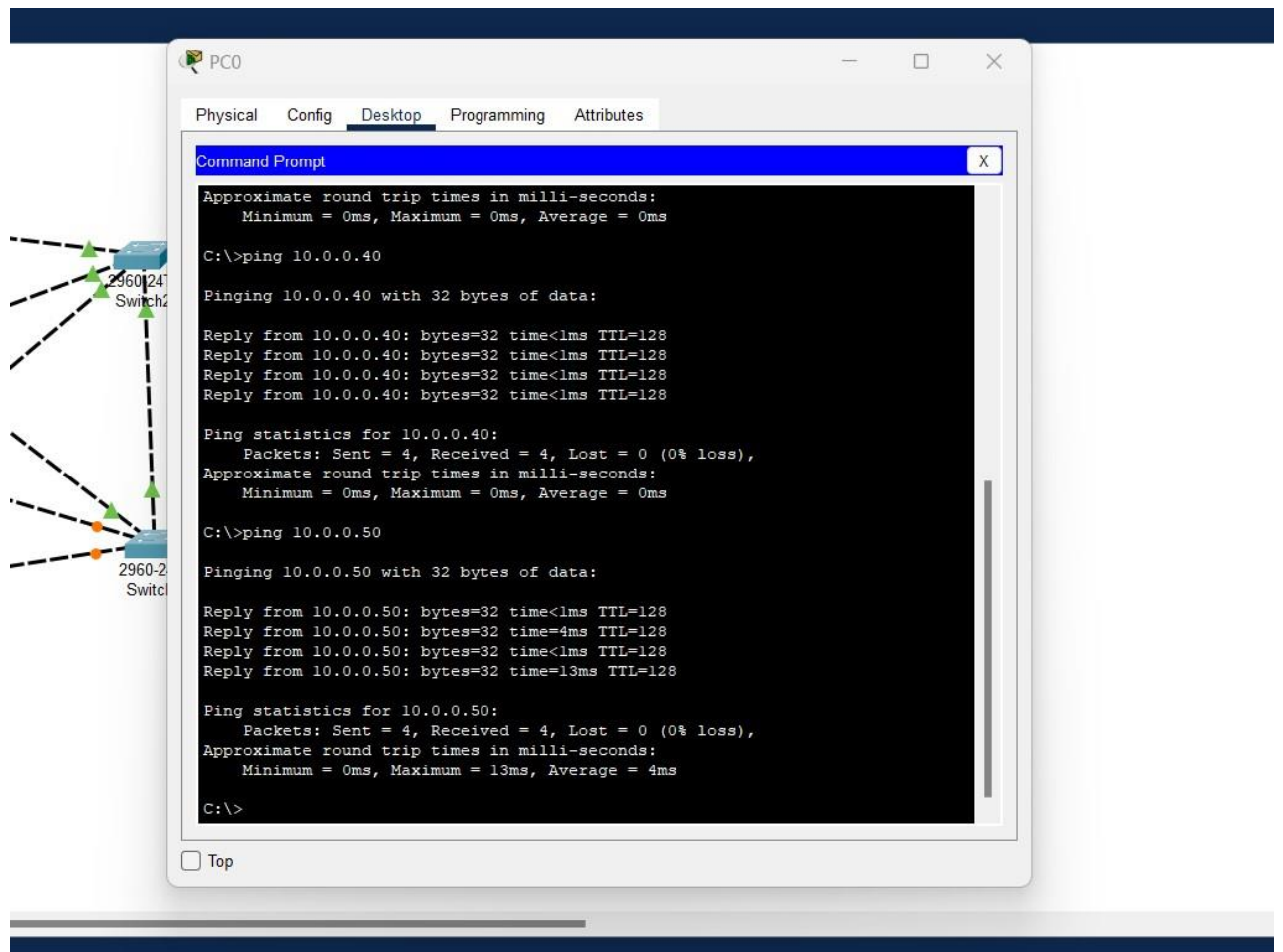
- According to the diagram (last page) use the Fa0/1 to Fa 0/4 in each Core Switches (5 Points)

Step 4 (20 Points):

Connect a PC to each switch's VLAN 35 (Students) interfaces (total 5 PCs) and make sure they can send ICMP messages (pings) to other PCs. The IP addresses and subnet masks have been noted on the Topology (below).

- From one of the PCs try to ping the other 4 PCs and put the **screenshot** here (10 Points)





Step 5 (20 Points): Hint: Use the command: Switch#show spanning-tree

Check the switches and find the Root Bridge for each VLAN then indicate the Switch hostname (as Root Bridge) on below table for each VLAN:

VLAN ID	1 (2 Points)	10 (3 Points)	15 (3 Points)	20 (3 Points)	25 (3 Points)	30 (3 Points)	35 (3 Points)
Root Bridge	32769	32778	32783	32788	32793	32798	32803

Submission:

Please submit your Packet Tracer file (PKT file) and this word file including your answers in the Assignment_1 link. **Do not compress (zip) the file.**

Topology:

